A note on Open AI's "A counterexample to Hadwiger's conjecture"

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October 11, 2026

1 Summary

In the note they construct a triangle-free graph G that has many 4-cycles, such that for the complement graph we has its independence number $\alpha(\bar{G}) = 2$ and connected matching number $\mathrm{cm}(\bar{G}) < m/100$ where m is number of vertices. The connected matching number is defined as follows:

Definition 1. *A matching M in a graph H is called a* connected matching *if every two distinct edges of M are joined by an edge of H: for all distinct $e, f \in M$, there exist $u \in V(e)$ and $v \in V(f)$ such that $uv \in E(H)$. The* connected matching number *is*

$$\mathrm{cm}(H) = \max\{|M| : M \text{ is a connected matching in } H\}.$$

The complement graph provides a counterexample to Hadwiger's conjecture, which asserts that $\chi(H) \leq h(H)$, where $h(H)$ is the largest integer t such that H contains a K_t minor. Indeed, Proposition 3.5 of the paper (proved in Appendix A) gives

$$h(H) \leq \frac{|V(H)| + 4\,\mathrm{cm}(H) + 2}{3}.$$

Applying this to $H = \bar{G}$, and using $\alpha(\bar{G}) \leq 2$ so that every color class has at most two vertices, we obtain, for m large enough,

$$h(\bar{G}) \leq \frac{m + 4\,\mathrm{cm}(\bar{G}) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \leq \chi(\bar{G}).$$

Thus $h(\bar{G}) < \chi(\bar{G})$, contrary to Hadwiger's conjecture.

2 The construction

2.1 Triangle-free graphs from linear algebra

We first describe the general linear-algebraic scheme and then specify the data used in the manuscript. All vector spaces, ranks, and matrix operations are over $\mathbb{F}_2$.

Let V, W be finite-dimensional vector spaces. Fix $a \in V^*$ and a symmetric bilinear form $T : V \times V \to \mathbb{F}_2$, and put

$$r(\lambda) = a + T(\lambda, \cdot) \in V^*.$$

Let Ω be the set of vertices. For each $o \in \Omega$, assign an injective linear map $U_o : V \to W$ and a linear functional $u_o \in W^*$. Then we can define the associated graph:

Definition 2. *Two vertices $i, j \in \Omega$ are adjacent if there exist $\lambda_i, \lambda_j \in V$ such that*

$$U_i\lambda_i = U_j\lambda_j, \tag{1}$$

$$u_j \circ U_i = r(\lambda_i), \qquad u_i \circ U_j = r(\lambda_j), \tag{2}$$

$$a(\lambda_i) + a(\lambda_j) = 1. \tag{3}$$

Equation (2) *is an equality in V^*.*

Lemma 1. *The associated graph is loopless and triangle-free.*

Proof. Loopless is guaranteed by injectivity of U and (3). Suppose i, j, k is a triangle. Write λ_{ij} for the λ_i at i for the pair ij. Evaluate (2) on λ_{ik}:

$$u_j(U_i\lambda_{ik}) = a(\lambda_{ik}) + T(\lambda_{ij}, \lambda_{ik}).$$

Then sum six equations by permuting i, j, k. For fixed j, the two terms on the left cancel because $U_i\lambda_{ik} = U_k\lambda_{ki}$. For fixed i, the two bilinear terms cancel by symmetry of T. The remaining right-hand side is

$$\sum_{\{i,j\}} \big(a(\lambda_{ij}) + a(\lambda_{ji})\big) = 1 + 1 + 1 = 1,$$

whereas the left-hand side is zero, a contradiction. □

2.2 Parameters

Starting from the number of vertices m that is large enough. The manuscript fixes

$$C_0 = 1000, \qquad g = 10^9 + 1, \qquad D = 4C_0 g, \qquad M = 2^{1000}.$$

And there are constants b, j_*, J, r_0, M_0 are chosen in Appendix A of the manuscript, independently of m but not sepcified.

Set

$$h = 1000r_0, \qquad N = M_0 n, \qquad m = 2^{C_0 g N}.$$

So we will use $n \to \infty$ as pivotal variable afterwards.

2.3 Boolean moment spaces

Let

$$I = \mathbb{Z}/g\mathbb{Z}, \quad \mathcal{E} = \binom{I}{2}, \quad I_d = \{d+1, \dots, d+(g-1)/2\} \subset I.$$

Elements of I are *tags* and elements of $\mathcal{E}$ are *components*. A *Boolean point* consists of a *selector* $s \in \mathbb{F}_2^b$ and *base coordinates* z in g^2+3 blocks of n bits:

$$z = \left((z_{O_{d,t}})_{d,t\in I},\ z_S,\ z_\#,\ z_Z\right) \in (\mathbb{F}_2^n)^{g^2+3}.$$

For each block label $A \in \{O_{d,t} : d, t \in I\} \cup \{S, \#, Z\}$, write

$$z_A = (z_{A,1}, \dots, z_{A,n}) \in \mathbb{F}_2^n.$$

Thus z has g^2+3 blocks, each containing n bits; $z_{A,k}$ denotes the kth bit in block A. Write

$$p_* = \sum_{j=0}^{j_*} \binom{b}{j}, \qquad B = \mathbb{F}_2^{p_*} \otimes \mathbb{F}_2^{1+(g^2+3)n}.$$

Let $p(s) \in \mathbb{F}_2^{p_*}$ be the bit vector listing the evaluations at s of all squarefree selector monomials of degree at most j_*, in a fixed order. Define the evaluation vector and its point moment by

$$v(s,z) = p(s) \otimes (1, z), \qquad w(s,z) = v(s,z) \otimes v(s,z).$$

Let $W \subseteq B \otimes B$ be the span of all point moments. For $\ell \in I$, let W_ℓ be the span of those point moments whose base values satisfy $O_{d,t} = 0$ whenever $\ell \notin I_d$. Let W_* be the span obtained by setting all O-blocks to zero.

Define the linear cut-profile map

$$\begin{aligned} f : \bigoplus_{\ell\in I} W_\ell &\longrightarrow \bigoplus_{e\in\mathcal{E}} W, \\ (w_\ell)_{\ell\in I} &\longmapsto \left(w_\ell + w_{\ell'}\right)_{\{\ell,\ell'\}\in\mathcal{E}}. \end{aligned}$$

The *witness space* is $X = \operatorname{im} f$. A tuple $(w_\ell)_{\ell\in I}$ is a representation of $x \in X$ if $f((w_\ell)_{\ell\in I}) = x$. By Lemma 2.3 of the manuscript,

$$\ker f = \{(w_*)_{\ell\in I} : w_* \in W_*\},$$

that is, two representations of the same x differ by a constant tuple with value in W_*.

2.4 The V, W, T and a

We can now identify the two vector spaces from Section 2.1. To distinguish their roles from the notation used in the concrete construction, write them as $V_{\S 2.1}$ and $W_{\S 2.1}$:

$$V_{\S 2.1} = X = \operatorname{im} f, \qquad W_{\S 2.1} = \bigoplus_{e \in \mathcal{E}} V_e^+ \otimes V_e^-, \qquad V_e^\pm = \mathbb{F}_2^N.$$

The first is the witness space: its elements are tuples of $d_B \times d_B$ matrices satisfying the cut-profile condition, where $d_B = \dim B$. The second is the ambient space: its elements are tuples of $N \times N$ matrices. A vertex o will determine an embedding

$$U_o : V_{\S 2.1} \longrightarrow W_{\S 2.1}, \qquad x \longmapsto (P_{o,e} x_e Q_{o,e}^T)_{e \in \mathcal{E}},$$

using the frame matrices defined in the next subsection.

Remark 1. *The $W_{\S 2.1}$ is not the moment space $W \subseteq B \otimes B$ defined above; the ambient space is denoted by V later in the concrete construction.*

So here we are ready to define $T \in Sym^2(X^*)$ and $a \in X^*$. Define *linear testers* on moments by selecting ordered matrix entries:

$$\eta_{d,t}(w(s,z)) = \sum_{\rho=1}^{r_0} z_{O_{d,t},2\rho-1} z_{O_{d,t},2\rho}, \; d,t \in I$$

$$\eta_S(w(s,z)) = \sum_{\rho=1}^{r_0} z_{S,2\rho-1} z_{S,2\rho}.$$

Put $\eta_O = \sum_{d,t} \eta_{d,t}$ and $\eta = \eta_O + \eta_S$. They should be understood as taking value in W^*.

For a cut profile x represented by (w_ℓ), set

$$a_t(x) = \sum_{\ell,d} \eta_{d,t}(w_\ell), \qquad a(x) = \sum_t a_t(x), \qquad b_t(x) = \sum_{\ell \neq t} \eta_S(w_\ell).$$

These functionals do not depend on the representation: the O-testers vanish on W_*, and a constant change contributes $(g-1)\eta_S(w_*) = 0$ to b_t.

Define

$$T_0(x,y) = a(x)a(y) + \sum_{t \in I} \big(a_t(x) b_t(y) + b_t(x) a_t(y)\big).$$

For each $1 \le j \le J$ and ordered pair $e, f \in \mathcal{E}$, fix endomorphisms L_j^{ef}, R_j^{ef} of B, chosen to satisfy the uniform mixing conditions of Lemma 5.5 of the manuscript. With the entrywise pairing $\langle A, B \rangle = \operatorname{Tr}(A^T B)$, put

$$B_1(x,y) = \sum_{j=1}^{J} \sum_{e,f \in \mathcal{E}} \left\langle x_e, L_j^{ef} y_f (R_j^{ef})^T \right\rangle,$$

$$T_1(x,y) = B_1(x,y) + B_1(y,x), \qquad T = T_0 + T_1.$$

Then T is symmetric and $T(x,x) = a(x)$. This specifies $r(\lambda) = a + T(\lambda, \cdot)$ in the edge definition.

The prescribed self-frame matrix E is also fixed. Let E_η be the sum of the ordered tester matrices, so $\langle E_\eta, w\rangle = \eta(w)$ for $w \in W$. Choose $n \times n$ matrices $M_1, \dots, M_b$ along with the mixers as in Lemma 5.5, and define $E_\#$ by

$$v(s,z)^T E_\# v(s',z') = \sum_{\rho=1}^{b} (s_\rho + s'_\rho) z_\#^T M_\rho z'_\#.$$

This is bilinear on B since $j_* \geq 1$. Set $E = E_\eta + E_\#$. Its additional term vanishes on coincident point inputs, so

$$\langle E, w\rangle = \eta(w) \qquad (w \in W).$$

The Z-block does not occur in E. The mixers and the matrices M_ρ are chosen once for each sufficiently large n, before graph sampling. Their existence with the required rank properties is supplied by Lemma 5.5; arbitrary choices need not give the connected-matching bound.

2.5 Random frames: the raw vertex law

For each $e \in \mathcal{E}$, take common ambient spaces $V_e^+ = V_e^- = \mathbb{F}_2^N$, paired by the dot product. Write $d_B = \dim B$. A raw vertex, also called an orientation, is a tuple

$$o = (P_{o,e}, Q_{o,e}, X_{o,e}, Y_{o,e})_{e \in \mathcal{E}},$$

where $P_{o,e}, Q_{o,e}$ are $N \times d_B$ matrices and $X_{o,e}, Y_{o,e}$ are $N \times h$ matrices. At each component sample uniformly among all quadruples satisfying

$$\begin{aligned} &[P_{o,e}\ X_{o,e}] \text{ and } [Q_{o,e}\ Y_{o,e}] \text{ have full column rank,} \\ &P_{o,e}^T Q_{o,e} = E, \\ &X_{o,e}^T Q_{o,e} = 0, \qquad P_{o,e}^T Y_{o,e} = 0. \end{aligned}$$

No value is prescribed for $X_{o,e}^T Y_{o,e}$. The frame set is nonempty when $N \geq 2(d_B + h)$, as ensured by the choice of M_0. Components are sampled independently; the four matrices within a component are constrained and hence need not be independent. Let Ω_n denote this finite raw vertex space and μ_n its law. The frame matrix $X_{o,e}$ is distinct from the witness space X.

Set

$$V = \bigoplus_{e \in \mathcal{E}} V_e^+ \otimes V_e^-.$$

Identifying tensors with matrices, define

$$U_o(x) = (P_{o,e} x_e Q_{o,e}^T)_{e\in\mathcal{E}}, \qquad u_o(\xi) = \sum_{e\in\mathcal{E}} \langle Y_{o,e} X_{o,e}^T, \xi_e \rangle.$$

The full-rank conditions make U_o injective: left inverses of $P_{o,e}$ and $Q_{o,e}$ recover x_e from its image. The two zero cross-Gram conditions imply $u_o \circ U_o = 0$. We have therefore supplied all the data needed for the general edge definition.

2.6 The sampled graph and the remaining theorem

Sample

$$o_1, \dots, o_m \overset{\text{iid}}{\sim} \mu_n, \qquad m = 2^{C_0 g N}.$$

On the positions $1, \dots, m$, put $ij \in E(G)$ exactly when o_i, o_j satisfy the edge criterion (1)–(3); put $ij \in E(H)$ otherwise. This completes the definition of the random graph. Its edges are determined by the sampled frames, with no further independent edge sampling.

Remark 2. *The initial random variable is the tuple of frame matrices o, not a Boolean point (s, z). The monomial list and moment spaces are fixed globally. Boolean points vary when constructing witnesses; sampling point parameters with the frames fixed is called a query in the manuscript.*

Triangle-freeness of G, and hence $\alpha(H) \le 2$, holds for every sample. The harder assertion is that $\mathrm{cm}(H) < m/100$ with probability $1 - \exp(-\Omega(m))$ (Proposition 3.4). To obtain it, the manuscript calls an ordered pair of raw vertices that are nonadjacent in G a *unit*; such a pair represents an edge of H when placed at distinct sampled positions. Two units conflict if all four cross pairs are edges of G, so no edge of H joins their endpoint sets. Their corresponding edges therefore cannot coexist in a connected matching.

Theorem 3.1 asserts that every law σ on units satisfying the pointwise density bounds

$$\sigma_1 \le M\mu_n, \qquad \sigma_2 \le M\mu_n, \qquad \sigma \le 2^{DN}(\mu_n \otimes \mu_n)$$

gives a conflict between two independent σ-units with probability at least 2^{-100gN}. The endpoints within one unit may be dependent. An entropy-based container argument transfers this statement to the sampled graph. Thus the construction above specifies the graph; the subsequent distribution theorem and container argument establish the small connected matching number needed for the counterexample.

A The clique-minor bound (Proposition 3.5)

Proposition 1. *For every finite nonempty graph H with m vertices,*

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3}.$$

Proof. (Idea) Write $c = \mathrm{cm}(H)$, $b = h(H)$. Thus that there are b pairwise disjoint, nonempty connected vertex sets, with an edge joining every pair of sets. These sets are the branch sets of a K_b minor. Let s be the number of sets with one vertex and e be the number of sets with two vertices.

Firstly $c > e + \frac{s-1}{2}$: Each two-vertex branch set has an edge. And pairs of singleton branches supports an edge, the set of them is a connected matching, so we have the desired inequality.

Then every remaining branch set has at least three vertices. Since the branch sets are disjoint,

$$m \geq s + 2e + 3(b - s - e) = 3b - 2s - e.$$

Combining two inequality give use the contradiction. □